WHERE VIOLENCE PERSISTS: LATENT SPATIAL FIELDS FOR THE RESIDUAL RISK OF ARMED CONFLICT AND SOCIAL UNREST

BY STEFANO CASTRUCCIO[1,a] AND RICHARD MARCANTONIO[2]

[1]*Department of Applied and Computational Mathematics and Statistics, University of Notre Dame,* [a]*scastruc@nd.edu*

[2]*Kroc Institute for International Peace Studies, Keough School of Global Affairs, University of Notre Dame*

[TODO: Abstract to be written after the application.]

1. Introduction. [TODO: Introduction (not drafted yet).]

2. Data.

2.1. *Conflict and unrest events.* Event data come from the Armed Conflict Location & Event Data Project (ACLED; Raleigh et al., 2010), which records the date, location, actors and type of individual political violence and demonstration events from news, local partners and other sources. We use events from 1 January 2018 to 31 December 2024 and group ACLED event types into two outcomes: *violent conflict* (battles, explosions/remote violence and violence against civilians) and *social unrest* (riots and protests). Events are aggregated to a regular $0.5^\circ \times 0.5^\circ$ longitude–latitude grid and to calendar weeks. For cell s and week t let $N_{st}^{(j)}$ be the number of events of type j and let

$$Y_{st} = \mathbf{1}\Big\{\textstyle\sum_{j\in\mathcal{J}} N_{st}^{(j)} > 0\Big\},$$

where $\mathcal{J}$ is the set of event types of the outcome being modelled; the two outcomes are analysed separately. Event weeks are rare: in the 2023–2024 period, 1.62% of grid-cell weeks contain at least one conflict event and 2.73% at least one unrest event.

The grid consists of all 0.5° cells that intersect a country polygon of Natural Earth (1:10m), excluding cells whose biome is rock and ice. Each cell is assigned to the single country that covers its centroid. National indicators are attached through that country only, so every cell-week appears exactly once. The result is 59,344 cells observed for 366 weeks, or 21,719,904 cell-weeks. The table produced by the earlier pipeline replicated every cell once for each country and IPC region it intersects, at every join of a national indicator, and has 70.0 million rows for 34.7 million distinct cell-weeks (before excluding rock and ice). Section 5.1 describes the reconstruction and its consequences.

Two features of ACLED matter for modelling. First, geographic coverage expanded during the study period, with some countries entering the dataset after 2018; cell weeks in a country before its coverage start are *unobserved*, not event free, and are removed rather than coded as zeros. [TODO: Add the coverage start date by country and the number of cell weeks removed.] Second, the probability that an event is recorded depends on media and partner density, so the outcome is the product of event occurrence and reporting. Neither is separately identified without auxiliary information; the latent fields introduced in Section 3 absorb spatially smooth reporting differences together with unmeasured drivers of risk, and we interpret them accordingly.

Keywords and phrases. armed conflict, ACLED, latent Gaussian models, SPDE, spatial confounding, spatially varying coefficients, gradient boosting.

2.2. *Recent event history.* For each cell and week we compute the number of events of each type in the four preceding weeks, $L_{st}^{(j)} = \sum_{l=1}^{4} N_{s,t-l}^{(j)}$, using only past weeks. In the models of Section 3 the history enters through $h_{st} = \log\bigl(1 + \sum_{j \in \mathcal{J}} L_{st}^{(j)}\bigr)$ for the outcome being modelled and, as additional covariates, the corresponding quantities for the other outcome.

2.3. *Environmental, socioeconomic and national covariates.* Covariates are listed in Table 1. They vary at three resolutions, which is central to the identification arguments of Section 3.6:

1. *cell–week*: four-week averages of daily minimum and maximum temperature, heat index and wet-bulb globe temperature (CHIRTS), precipitation (CHIRPS), $PM_{2.5}$ (CAMS) and a drought index (Global Drought Observatory);
2. *cell*: population (GHSL), river length and lake area (Natural Earth), malaria risk (Malaria Atlas Project), terrestrial biome (RESOLVE Ecoregions) and distance from the equator;
3. *country–year*: GDP per capita (apportioned to cells by population share), access to electricity, the Environmental Performance Index (EPI), the V-Dem liberal democracy index, and per-capita greenhouse gas emissions, material footprint and ecological footprint.

Food insecurity (IPC phases) is available at the subnational level and is apportioned to cells by population share.

Because a raw temperature means different things in different climates, the climate covariates enter the statistical models as standardised anomalies with respect to the cell- and calendar-week-specific 2018–2024 climatology,

$$z_{st} = \bigl(x_{st} - \bar{x}_{s,\mathrm{wk}(t)}\bigr) / \mathrm{sd}_{s,\mathrm{wk}(t)},$$

so that their coefficients are identified from within-cell variation over time and are not confounded by the persistent spatial pattern of the climatology itself. The climatology enters separately as a cell-level covariate. [TODO: Decide whether climatology uses a longer reference period (e.g. 1991–2020) from CHIRTS/CHIRPS.]

2.4. *Exploratory evidence of residual spatial dependence.* Residuals of a spatially cross-fitted XGBoost model with the covariates above and recent event history are strongly spatially dependent for conflict. In Africa and the Middle East, Moran's I of the standardised cell residuals is 0.45, the semivariogram rises over the first 300–400 km, and the residual correlation between neighbouring cells is larger within countries (0.48) than across borders (0.31). For unrest the dependence is weaker (Moran's I 0.11) and local. These diagnostics, described in Section 3.9 and reported in Section 5.3, motivate the latent fields of Section 3. [TODO: Add a map of event-week frequencies by outcome and the residual temporal autocorrelation.]

3. Methods.

3.1. *A latent Gaussian model for event occurrence.* Let $\mathcal{S}$ be the set of n grid cells, $c(s)$ the country and $b(s)$ the biome of cell s, and $t = 1, \dots, T$ the weeks. Conditionally on a set of latent effects, the outcomes are independent Bernoulli variables, $Y_{st} \mid \pi_{st} \sim \mathrm{Bernoulli}(\pi_{st})$, with

$$\mathrm{logit}\, \pi_{st} = \boldsymbol{x}_{st}^{\top}\boldsymbol{\beta} + \delta\, h_{st} + \sum_{k \in \mathcal{K}} z_{k,st}\, \beta_k(s) + \boldsymbol{z}_{c(s)}^{\top}\boldsymbol{\alpha} + u_{c(s)} + w(s) + v(s,t). \tag{3.1}$$

Here $\boldsymbol{x}_{st}$ collects cell and cell–week covariates with fixed effects; h_{st} is the recent event history of Section 2.2, so δ measures *state dependence*; $z_{k,st}$, $k \in \mathcal{K}$, are the climate anomalies,

TABLE 1
Covariates, resolution and source.

Covariate	Resolution	Source
Minimum / maximum temperature (4-week mean)	cell–week	CHIRTS
Heat index, wet-bulb globe temperature	cell–week	CHIRTS
Precipitation (4-week mean)	cell–week	CHIRPS
$PM_{2.5}$	cell–week	CAMS
Drought index	cell–week	Global Drought Observatory
Events of each type in previous 4 weeks	cell–week	ACLED
Population	cell	GHSL
River length, lake area	cell	Natural Earth
Malaria risk	cell	Malaria Atlas Project
Biome	cell	RESOLVE Ecoregions
Distance from the equator	cell	derived
Food insecurity (IPC phases 1–5)	subnational–period	IPC
GDP per capita	country–year	World Bank
Access to electricity	country–year	World Bank
Environmental Performance Index	country–year	Yale EPI
Liberal democracy index	country–year	V-Dem
GHG emissions, material and ecological footprint p.c.	country–year	OWID, GFN

TABLE 2
Latent blocks of model (3.1), their precision matrices and hyperparameters.

Block	Dimension	Precision	Hyperparameters
fixed effects $\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}$	p	$10^{-4}\boldsymbol{I}$	–
country effects $\boldsymbol{u}$	n_c	$\sigma_u^{-2}\boldsymbol{I}$	σ_u
spatial field $\boldsymbol{w}$	n	$\boldsymbol{Q}_s(\rho_w, \sigma_w)$, eq. (3.3)	ρ_w, σ_w
SVC fields $\boldsymbol{\eta}_k$	n each	$\boldsymbol{Q}_s(\rho_k, \sigma_k)$	ρ_k, σ_k
space-time field $\boldsymbol{v}$	nT_v	$\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$	ρ_v, σ_v, ϕ

whose effects $\beta_k(s)$ vary in space (Section 3.5); $\boldsymbol{z}_c$ are national indicators with effects $\boldsymbol{\alpha}$; $u_c \sim \mathcal{N}(0, \sigma_u^2)$ are independent country effects; w is a stationary Matérn field on the sphere (Section 3.2); and v is a space-time field (Section 3.3). The three latent terms represent persistent risk at three scales: national institutions and reporting regimes (u), regional hotspots that cross borders (w), and slowly evolving episodes (v).

Model (3.1) is a latent Gaussian model (Rue, Martino and Chopin, 2009): stacking all latent quantities into $\boldsymbol{\xi} = (\boldsymbol{\beta}, \delta, \boldsymbol{\gamma}, \boldsymbol{\alpha}, \boldsymbol{u}, \boldsymbol{w}, \boldsymbol{\eta}, \boldsymbol{v})$, the linear predictor is $\boldsymbol{\eta} = \boldsymbol{o} + \boldsymbol{A}\boldsymbol{\xi}$ for a sparse design matrix $\boldsymbol{A}$ and an optional offset $\boldsymbol{o}$, and $\boldsymbol{\xi} \mid \boldsymbol{\theta} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}(\boldsymbol{\theta})^{-1}\}$ with a sparse, block-diagonal precision whose blocks are listed in Table 2. Fixed effects receive vague $\mathcal{N}(0, 10^4)$ priors and are integrated out together with the random effects.

3.2. *Matérn fields on the sphere via a finite-volume SPDE.* Following Lindgren, Rue and Lindström (2011), a Gaussian field w on $\mathbb{S}^2$ with Matérn-like covariance is defined as the stationary solution of

$$\tau\,(\kappa^2 - \Delta_{\mathbb{S}^2})\,w(s) = \mathcal{W}(s), \qquad s \in \mathbb{S}^2, \tag{3.2}$$

where $\Delta_{\mathbb{S}^2}$ is the Laplace–Beltrami operator and $\mathcal{W}$ is Gaussian white noise. This is the $\alpha = 2$ case, corresponding to Matérn smoothness $\nu = 1$ in the planar limit, with practical range $\rho = \sqrt{8}/\kappa$ and marginal variance $\sigma^2 = 1/(4\pi\kappa^2\tau^2)$ when ρ is small relative to the Earth's radius R. We parameterise the field by (ρ, σ) in kilometres and logit units.

The grid cells are natural control volumes, so instead of a triangulated mesh we discretise (3.2) by finite volumes on the longitude–latitude lattice itself. Integrating (3.2) over cell A_i and applying the divergence theorem with a two-point flux approximation gives

$$\tau\Big\{\kappa^2 a_i w_i + \sum_{j\sim i} \frac{\ell_{ij}}{d_{ij}}(w_i - w_j)\Big\} = \mathcal{W}(A_i) \sim \mathcal{N}(0, a_i),$$

where a_i is the area of cell i, $j \sim i$ ranges over the cells sharing an edge with i, ℓ_{ij} is the length of the shared edge and d_{ij} the great-circle distance between the centres. For a grid with spacings $\Delta\lambda, \Delta\varphi$ and centre latitude φ_i,

$$\begin{aligned} a_i &= R^2 \Delta\lambda \,\{\sin(\varphi_i + \Delta\varphi/2) - \sin(\varphi_i - \Delta\varphi/2)\}, \\ \ell_{ij}/d_{ij} &= \Delta\varphi/(\cos\varphi_i\, \Delta\lambda) \quad \text{(east–west neighbours)}, \\ \ell_{ij}/d_{ij} &= \cos\varphi_{ij}\, \Delta\lambda/\Delta\varphi \quad \text{(north–south neighbours)}, \end{aligned}$$

with φ_{ij} the latitude of the shared edge. Writing $\boldsymbol{C} = \mathrm{diag}(a_i)$ and $\boldsymbol{G}$ for the weighted graph Laplacian with edge weights ℓ_{ij}/d_{ij}, we get $\tau \boldsymbol{K}\boldsymbol{w} = \boldsymbol{C}^{1/2}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \kappa^2\boldsymbol{C} + \boldsymbol{G}$ and $\boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$, so

(3.3)

$$\boldsymbol{w} \sim \mathcal{N}\{\boldsymbol{0}, \boldsymbol{Q}_s^{-1}\}, \qquad \boldsymbol{Q}_s(\rho,\sigma) = \tau^2 \boldsymbol{K}\boldsymbol{C}^{-1}\boldsymbol{K}, \qquad \log|\boldsymbol{Q}_s| = 2n\log\tau + 2\log|\boldsymbol{K}| - \textstyle\sum_i \log a_i.$$

$\boldsymbol{Q}_s$ has at most 13 non-zeros per row. The construction is the lumped-mass finite-volume analogue of the finite-element representation in Lindgren, Rue and Lindström (2011). Because it uses the physical cell geometry, it accounts for the shrinking east–west spacing towards the poles, can be restricted to land cells by deleting rows and columns, and wraps around in longitude for global analyses.

REMARK 1. The discretisation imposes reflecting (Neumann) conditions on the boundary of the lattice, which roughly doubles the marginal variance of boundary cells (Section 4.2). For the global land grid the boundaries are the coastlines. Extending the lattice into the sea removes the inflation but lets correlation leak across narrow seas, for example between Yemen and Somalia across the Gulf of Aden, which is substantively wrong for conflict risk. We therefore use the barrier model of Bakka et al. (2019): the SPDE is written as

$$u(s) - \nabla\cdot\Big\{\tfrac{r(s)^2}{8}\nabla u(s)\Big\} = r(s)\sqrt{\pi/2}\,\sigma\,\mathcal{W}(s),$$

which reduces to (3.2) with $\kappa = \sqrt{8}/\rho$ when $r(s) \equiv \rho$, and sea cells within a buffer of the coast get range $r(s) = 0.1\rho$. The finite-volume form is $\boldsymbol{K}\boldsymbol{u} = \boldsymbol{D}\boldsymbol{\epsilon}$ with $\boldsymbol{K} = \boldsymbol{C} + (\rho^2/8)\boldsymbol{G}_m$, where $\boldsymbol{G}_m$ has edge weights $\tfrac{1}{2}(m_i + m_j)\ell_{ij}/d_{ij}$, $m_i = r_i^2/\rho^2$, and $\boldsymbol{D} = \mathrm{diag}\{r_i\sigma(\pi a_i/2)^{1/2}\}$, so that $\boldsymbol{Q}_s = \boldsymbol{K}\boldsymbol{D}^{-2}\boldsymbol{K}$. Sea cells carry no data. On a test domain with a 330 km strait, the barrier reduces the coastal variance from 2.02 to 1.31 (nominal 1) and the correlation across the strait from 0.14 (sea treated as land) to 0.01.

3.2.0.1. Priors.. We use penalised-complexity priors (Simpson et al., 2017; Fuglstad et al., 2019): $P(\rho < \rho_0) = 0.05$ and $P(\sigma > \sigma_0) = 0.01$, with $\rho_0 = 100$ km (twice the grid spacing) and $\sigma_0 = 1$, so that a field moving the odds of an event by a factor $e^{\pm 2}$ across two standard deviations is a priori unlikely. The country effect has a PC prior with $P(\sigma_u > 1) = 0.01$.

3.3. *Space-time persistence.* The space-time term is a separable field that evolves as a first-order autoregression of Matérn innovations,

$$\boldsymbol{v}_\tau = \phi\, \boldsymbol{v}_{\tau-1} + \sqrt{1-\phi^2}\, \boldsymbol{\epsilon}_\tau, \qquad \boldsymbol{\epsilon}_\tau \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_v, \sigma_v)^{-1}\},$$

(Cameletti et al., 2013), so that $\boldsymbol{v}$ has precision $\boldsymbol{Q}_{\mathrm{AR}}(\phi) \otimes \boldsymbol{Q}_s(\rho_v, \sigma_v)$, where $\boldsymbol{Q}_{\mathrm{AR}}$ is the tridiagonal precision of a unit-variance AR(1) process. To control cost, τ indexes blocks of Δ weeks ($\Delta = 4$ in the simulations), and $v(s,t) = v_{s,\lceil t/\Delta \rceil}$. The prior on $\operatorname{atanh}\phi$ is $\mathcal{N}(0, 1.5^2)$.

3.4. *State dependence versus persistent heterogeneity.* The finding that recent event counts dominate the importance rankings of boosted trees (Menicali, Marcantonio and Castruccio, 2026) has two very different explanations. Under *state dependence* ($\delta > 0$), an event raises the probability of future events in the same place, for example through retaliation, mobilisation or the arrival of armed groups. Under *persistent heterogeneity*, some places are simply riskier (w and v are large there), and past events are informative only as a proxy for that latent risk (Heckman, 1981). Both produce strong autocorrelation in Y_{st} and a large predictive contribution of h_{st}. Only a model that includes both the lag and the latent persistence can tell them apart. For prediction, the distinction matters little. For policy it is central: under state dependence, preventing an event reduces future violence; under heterogeneity, it does not.

In model (3.1), δ is identified from the within-cell timing of events relative to the slowly varying $w + v$. We condition on the first four weeks of each cell, so the usual initial-conditions problem of dynamic random-effects models (Wooldridge, 2005) induces a bias of order $1/T$, which is negligible for $T \geq 100$ weeks.

3.5. *Spatially varying climate effects.* To assess whether climate effects differ across biomes and how much variation remains within them, the effect of anomaly k is decomposed as

$$\beta_k(s) = \gamma_{k,b(s)} + \eta_k(s), \qquad \boldsymbol{\eta}_k \sim \mathcal{N}\{\mathbf{0}, \boldsymbol{Q}_s(\rho_k, \sigma_k)^{-1}\}, \tag{3.4}$$

a spatially varying coefficient model (Gelfand et al., 2003) whose mean is pooled by biome. The biome means $\gamma_{k,b}$ are fixed effects, and σ_k measures heterogeneity within biomes, with $\sigma_k = 0$ meaning that biome fully explains the variation. The design of $\boldsymbol{\eta}_k$ is the sparse matrix with entry $z_{k,st}$ in row (s,t) and column s.

Two estimands should be distinguished. The *biome-average effect* $\bar{\beta}_{k,b} = |\mathcal{S}_b|^{-1} \sum_{s \in \mathcal{S}_b} \beta_k(s)$, the average over the cells of biome b, is what a model with fixed biome-specific slopes and no $\boldsymbol{\eta}_k$ estimates. The *biome mean* $\gamma_{k,b}$ is the mean of the process that generated those cells, and it is what (3.4) estimates, with uncertainty that accounts for the finite number of independent regions in each biome. Both differ from the summary used in the SHAP-based analysis of Menicali, Marcantonio and Castruccio (2026), the median SHAP value within a biome. For a feature with an additive effect, that summary is $\operatorname{median}_{s \in \mathcal{S}_b}\{\beta_k(s)(z_{k,st} - \mathbb{E} z_k)\}$, which reflects where the biome's covariate values lie relative to the global mean as much as the strength of the effect. With anomalies centred at zero in every cell, it is nearly constant across biomes and unrelated to the effect (Section 4.6).

3.6. *National indicators and spatial confounding.* National indicators $\boldsymbol{z}_c$ such as the EPI or the ecological footprint are constant within a country and year. With n_c countries their effects $\boldsymbol{\alpha}$ are identified only from between-country contrasts, so the effective sample size for $\boldsymbol{\alpha}$ is of order n_c, not the number of cell weeks. Ignoring this, by modelling cell weeks as conditionally independent or bootstrapping blocks that cut across countries, produces intervals that are far too narrow. More importantly, if the indicators are associated with unobserved

country-level risk, so that $\mathbb{E}\{u_c + \bar{w}_c \mid \boldsymbol{z}_c\} \neq 0$ with $\bar{w}_c$ the country average of w, then $\boldsymbol{\alpha}$ is confounded. An indicator of environmental performance that tracks state capacity is the leading example. This is the spatial confounding problem (Reich, Hodges and Zadnik, 2006; Hodges and Reich, 2010; Paciorek, 2010): adding a spatial random effect changes the estimate of a spatially smooth covariate's coefficient, and neither the estimate with the random effect nor the one without it is unbiased for a causal effect (Khan and Calder, 2022; Zimmerman and Ver Hoef, 2022).

We report three estimators and treat their disagreement as a measure of how fragile the national-level conclusions are: (i) the full model (3.1) with $\boldsymbol{u}$ and $\boldsymbol{w}$; (ii) the same model without $\boldsymbol{u}$ and $\boldsymbol{w}$; (iii) a *spatial+* estimator (Dupont, Wood and Augustin, 2022), in which $\boldsymbol{z}_c$ is replaced by its residual after projection on the m smoothest eigenvectors of the generalised eigenproblem $\boldsymbol{G}\boldsymbol{e} = \lambda \boldsymbol{C}\boldsymbol{e}$. These are the GMRF analogue of the thin-plate eigenvectors of Dupont, Wood and Augustin (2022), so the residual retains only the variation of $\boldsymbol{z}_c$ that is too rough to be explained by the spatial field. The restricted spatial regression of Reich, Hodges and Zadnik (2006) is not used, because its intervals are known to be anticonservative (Hanks et al., 2015; Khan and Calder, 2022).

3.7. *Hybrid models: boosting offsets with a residual field.* To retain the flexibility of gradient boosting for the many weakly informative covariates, we also fit a semiparametric version of (3.1) in which the fixed part is replaced by the logit $\hat{f}(\boldsymbol{x}_{st})$ of an XGBoost classifier (Chen and Guestrin, 2016) and enters as an offset,

$$\operatorname{logit} \pi_{st} = \hat{f}^{(-k(s))}(\boldsymbol{x}_{st}) + \beta_0 + u_{c(s)} + w(s)\ [+ v(s,t)]. \tag{3.5}$$

The offset must be *spatially cross-fitted*. Cells are grouped into $5^\circ \times 5^\circ$ blocks, the blocks are assigned at random to $K = 5$ folds, and $\hat{f}^{(-k)}$ is trained on the training-period rows of all folds except k. It then provides the offset for every row of fold k, in both the training and test periods. Without cross-fitting, trees that receive national indicators can identify countries, and through the interaction of population, biome and distance from the equator, approximate locations. In-sample boosting residuals therefore understate the spatial structure, and adding a field on top double counts it. The construction is the spatial analogue of cross-fitting in double machine learning (Chernozhukov et al., 2018). It differs from GPBoost (Sigrist, 2022), which alternates boosting steps with Gaussian process updates, in that the boosting fit is fixed before the latent model is fitted. This makes model (3.5) cheap enough for the global data and allows the same Laplace machinery to be used.

The boosting analysis of Menicali, Marcantonio and Castruccio (2026) trained XGBoost with positives weighted by $\omega = (n_0/n_1)^{1/2}$. Under the logistic loss this shifts the fitted log-odds by $\log \omega$ (King and Zeng, 2001), so offsets are prior-corrected, $\hat{f} \leftarrow \hat{f} - \log \omega$, or the boosting model is retrained without weights.

3.8. *Inference.* For fixed hyperparameters $\boldsymbol{\theta}$, the posterior mode $\boldsymbol{\xi}^* = \arg\max_{\boldsymbol{\xi}} \{\log p(\boldsymbol{y} \mid \boldsymbol{\xi}) - \tfrac{1}{2}\boldsymbol{\xi}^\top \boldsymbol{Q}\boldsymbol{\xi}\}$ is found by damped Newton iterations,

$$\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} + \lambda\, (\boldsymbol{Q} + \boldsymbol{A}^\top \boldsymbol{W} \boldsymbol{A})^{-1} \{\boldsymbol{A}^\top (\boldsymbol{y} - \boldsymbol{\pi}) - \boldsymbol{Q}\boldsymbol{\xi}\}, \qquad \boldsymbol{W} = \operatorname{diag}\{\pi_{st}(1 - \pi_{st})\},$$

with step halving. The Laplace approximation to the marginal likelihood is

$$\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) = \log p(\boldsymbol{y} \mid \boldsymbol{\xi}^*) - \tfrac{1}{2}\boldsymbol{\xi}^{*\top} \boldsymbol{Q}\boldsymbol{\xi}^* + \tfrac{1}{2} \log |\boldsymbol{Q}| - \tfrac{1}{2} \log |\boldsymbol{Q} + \boldsymbol{H}|, \qquad \boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}^* \boldsymbol{A}. \tag{3.6}$$

The hyperparameters are set to $\hat{\boldsymbol{\theta}} = \arg\max \{\log \tilde{p}(\boldsymbol{y} \mid \boldsymbol{\theta}) + \log p(\boldsymbol{\theta})\}$ by L-BFGS-B on the internal scale (logarithms of ranges and standard deviations, $\operatorname{atanh} \phi$). Each evaluation warm-starts Newton at the previous mode. Given $\hat{\boldsymbol{\theta}}$, the latent posterior is approximated by

$\mathcal{N}\{\boldsymbol{\xi}^*, (\boldsymbol{Q}+\boldsymbol{H})^{-1}\}$. Samples are drawn without forming a Cholesky factor of $\boldsymbol{Q}+\boldsymbol{H}$ by the perturbation identity

$$\boldsymbol{\xi} = \boldsymbol{\xi}^* + (\boldsymbol{Q}+\boldsymbol{H})^{-1}\big(\boldsymbol{R}^\top \boldsymbol{\epsilon}_1 + \boldsymbol{A}^\top \boldsymbol{W}^{*1/2}\boldsymbol{\epsilon}_2\big),$$

where $\boldsymbol{R}^\top\boldsymbol{R} = \boldsymbol{Q}$ is block-wise explicit, for instance $\boldsymbol{R} = \tau \boldsymbol{C}^{-1/2}\boldsymbol{K}$ for a Matérn block and a Kronecker product of a bidiagonal AR(1) factor with that matrix for $\boldsymbol{v}$. This is an empirical Bayes version of INLA (Rue, Martino and Chopin, 2009) and is similar in spirit to TMB (Kristensen et al., 2016). Integrating over $\boldsymbol{\theta}$ with a central composite design would widen the intervals slightly; the simulations of Section 4.2 show that plug-in intervals already have close to nominal coverage for the fixed effects and the field.

3.8.0.1. Computation.. All matrices are sparse. Every Newton step and every log-determinant requires a sparse factorisation of $\boldsymbol{Q}+\boldsymbol{H}$, whose sparsity pattern does not depend on $\boldsymbol{\theta}$. We therefore compute a nested-dissection ordering (Karypis and Kumar, 1998) once and reuse it. For the space-time model this reduced factorisation time by a factor of 7 to 14 compared with approximate minimum degree. Since $\boldsymbol{H} = \boldsymbol{A}^\top \boldsymbol{W}\boldsymbol{A}$ aggregates over weeks, the cost of the spatial and SVC blocks is driven by the number of cells, not by the number of cell weeks. A global 0.5° land grid gives $n \approx 6.5 \times 10^4$ latent values per field, comparable to the larger problems routinely handled by sparse Cholesky methods.

3.8.0.2. Prediction.. For a new cell week with design row $\boldsymbol{a}$ and offset o, the posterior predictive probability is $\mathbb{E}\{\operatorname{expit}(o + \boldsymbol{a}^\top\boldsymbol{\xi})\} \approx \operatorname{expit}\{m/\sqrt{1+\pi s^2/8}\}$, where m and s^2 are the posterior mean and variance of $o + \boldsymbol{a}^\top\boldsymbol{\xi}$; s^2 is estimated from posterior samples.

3.9. *Diagnostics and evaluation.*

3.9.0.1. Residual diagnostics.. Binary residuals are uninformative one at a time, so we aggregate over weeks within cells. The standardised cell residual

$$r_s = \sum_t (y_{st} - \hat{\pi}_{st}) \Big/ \Big\{\sum_t \hat{\pi}_{st}(1-\hat{\pi}_{st})\Big\}^{1/2}$$

is approximately standard normal and independent across cells when the model is correct and the outcomes are conditionally independent. We report Moran's I (Moran, 1950) of $\{r_s\}$ for rook adjacency with a permutation p-value, the empirical semivariogram of $\{r_s\}$ against great-circle distance, and the correlation of r_s between neighbouring cells in the same country versus across a border. A much larger within-country correlation points to missing country-level effects. Temporal dependence is assessed by the pooled within-cell autocorrelation of randomised quantile residuals (Dunn and Smyth, 1996).

3.9.0.2. Out-of-sample evaluation.. Models are trained on 2018–2022 and evaluated on 2023–2024, matching the original analysis. We report the logarithmic score and the Brier score, both strictly proper (Gneiting and Raftery, 2007), as well as the area under the precision–recall curve (average precision) and the ROC curve. Since prediction in the same places is dominated by persistence, every model is compared with a persistence baseline that ranks cell weeks by the number of events in the previous four weeks. To assess transferability to new places, we also evaluate with leave-block-out cross-validation over 5° blocks (Roberts et al., 2017).

4. Simulation studies.

We use simulation to answer five questions that the application raises but cannot settle on its own: (S0) whether the finite-volume SPDE and the empirical Bayes Laplace approximation recover the field and give calibrated intervals; (S1) whether residual spatial structure left by a boosting model can be detected and modelled, and whether

modelling it improves forecasts; (S2) how national indicators behave under spatial confounding; (S3) whether state dependence can be separated from persistent heterogeneity; (S4) whether biome-specific climate effects are recovered better by spatially varying coefficients than by SHAP summaries. All code is in the supplementary material.

4.1. *Data-generating process.* The simulator mimics the structure of the ACLED grid. The domain is a 40×30 block of 0.5° cells between 10°E and 30°E and between 5°S and 10°N (1,200 cells of about 55 km), with 104 training weeks, 52 test weeks and 12 burn-in weeks. Countries are the Voronoi cells of randomly placed seeds (14 countries; 25 in S2). Four biomes are latitude bands with boundaries perturbed by a smooth random field, ordered from "cold" (biome 1) to "tropical" (biome 4). Population is a standardised Matérn field (range 250 km) plus white noise. Temperature and precipitation anomalies are standardised AR(1) processes in time (coefficients 0.6 and 0.3) of Matérn fields with ranges of 800 and 500 km. Three national covariates Z_0, Z_1, Z_2 are standard normal across countries; Z_1 and Z_2 have no effect but, like the national indicators in the data, allow trees to identify countries. Outcomes are generated sequentially from

(4.1)
$$\operatorname{logit} \pi_{st} = \beta_0 + \beta_{\mathrm{pop}} \mathrm{pop}_s + \delta \log\Big(1 + \textstyle\sum_{l=1}^{4} y_{s,t-l}\Big) + \beta(s)\, \mathrm{temp}_{st} + \beta_{\mathrm{prec}} \mathrm{prec}_{st} + \alpha Z_{0,c(s)} + u_{c(s)} + w(s) + v(s,t),$$

with $\beta(s) = \gamma_{b(s)} + \eta(s)$, $u_c \sim \mathcal{N}(0, 0.3^2)$, and w, η and v drawn from the GMRFs of Section 3. Unless stated otherwise, $\beta_0 = -4$, $\beta_{\mathrm{pop}} = 0.5$, $\delta = 0.6$, $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$, $\eta \equiv 0$, $\beta_{\mathrm{prec}} = -0.15$, $\alpha = 0$, $\sigma_w = 1$, $\rho_w = 400$ km and $v \equiv 0$. The resulting prevalence of event weeks is 3–4%, close to that of the data. Spatial confounding (S2) is induced by setting $Z_{0,c} = \varrho\, \tilde{\ell}_c + (1 - \varrho^2)^{1/2} \epsilon_c$, where $\tilde{\ell}_c$ is the standardised country average of $u_c + w(s)$ and $\epsilon_c \sim \mathcal{N}(0, 1)$.

The boosting model is XGBoost with the hyperparameters of the original analysis (Menicali, Marcantonio and Castruccio, 2026) (depth 5, learning rate 0.05, row and column subsampling 0.8, minimum child weight 30, L_2 penalty 5; 300 rounds), without class weights. Its features are population, distance from the equator, biome, the two anomalies, Z_0–Z_2 and the four-week event count. As in the original analysis, the features include no coordinates.

4.2. *S0: accuracy of the field representation and of the Laplace engine.* We simulate w from the lattice SPDE (3.3) on the 40×30 domain with $\rho_w = 400$ km and $\sigma_w = 1$, and Bernoulli outcomes for 40 weeks from $\operatorname{logit} \pi_{st} = -2 + 0.5\, x_{st} + w(s)$ with $x_{st} \sim \mathcal{N}(0, 1)$, 50 replicates. Table 3 summarises the results. Across 2,000 draws, the empirical marginal variance of the field is 1.07 in cells at least eight cells (about 440 km) from the edge of the domain, against a nominal 1. At the edge it is 2.02, the doubling induced by the Neumann boundary that motivates the buffer of Remark 1. The hyperparameters are recovered with little bias ($\hat{\rho}_w = 399$ km, $\hat{\sigma}_w = 0.99$), and the posterior mean of w has correlation 0.96 with the truth. Plug-in 95% intervals cover 0.5 in 92% of replicates (Monte Carlo SE 3.8%) and cover $-2 + w(s)$ for 94.4% of cells, so ignoring the uncertainty in $\boldsymbol{\theta}$ costs little here. A fit takes about 5 seconds on one core.

4.3. *S1: residual structure after boosting.* Data are generated from (4.1) with the default parameters, and the models are trained on the first 104 weeks and evaluated on the following 52, as in the original analysis. Table 4 reports out-of-sample scores over 20 replicates and Table 5 the residual diagnostics and the recovery of the field. Figure 1 shows one replicate.

Three findings stand out. First, boosting does not absorb the residual field even with in-sample training on the same locations. Moran's I of the cell residuals of the XGBoost model as used in the original analysis is 0.33 (SD 0.07 across replicates), against an expected value

TABLE 3
S0: accuracy of the finite-volume SPDE field and of the empirical Bayes Laplace approximation (50 replicates).

Quantity	Mean	SD
$\hat{\rho}_w$ (km) [truth 400]	398.6	38.1
$\hat{\sigma}_w$ [truth 1]	0.985	0.075
$\hat{b}_1$ [truth 0.5]	0.491	0.013
Coverage of 95% interval for b_1	0.920	0.274
Coverage of 95% intervals for $b_0 + w(s)$ (share of cells)	0.944	0.008
$\mathrm{cor}(\hat{w}, w)$	0.956	0.008
Fit time (s)	5.6	0.9
Marginal variance of w, interior cells [nominal 1]	1.07	–
Marginal variance of w, edge cells	2.02	–

TABLE 4
S1: out-of-sample scores on the 52 test weeks (mean over 20 replicates, Monte Carlo standard error in parentheses). Lower is better for the log and Brier scores.

Model	Log score	Brier	AUC-PR	AUC-ROC
True probabilities	0.1429 (0.0080)	0.0367 (0.0026)	0.267 (0.018)	0.826 (0.005)
XGBoost (as in original analysis)	0.1501 (0.0084)	0.0377 (0.0027)	0.235 (0.017)	0.790 (0.006)
XGBoost, spatially cross-fitted	0.1649 (0.0098)	0.0402 (0.0031)	0.152 (0.015)	0.715 (0.008)
Cross-fitted XGBoost + w	0.1470 (0.0082)	0.0374 (0.0027)	0.245 (0.017)	0.809 (0.005)
Cross-fitted XGBoost + $u + w$	0.1469 (0.0082)	0.0374 (0.0027)	0.245 (0.017)	0.810 (0.005)
Parametric model with $u + w$	0.1457 (0.0081)	0.0371 (0.0026)	0.253 (0.018)	0.814 (0.005)
Persistence baseline	–	–	0.103 (0.009)	–

TABLE 5
S1: residual diagnostics and recovery of the field (mean and SD over 20 replicates).

Quantity	Mean	SD
Moran's I of cell residuals, XGBoost (original)	0.330	0.069
Moran's I of cell residuals, XGBoost cross-fitted	0.591	0.064
$\mathrm{cor}(\hat{w}, w)$, XGBoost + w	0.768	0.049
$\mathrm{cor}(\hat{w}, w)$, XGBoost + $u + w$	0.814	0.036
$\hat{\rho}_w$ (km), XGBoost + $u + w$ [truth 400]	373.374	48.288
$\hat{\sigma}_w$, XGBoost + $u + w$ [truth 1]	0.943	0.095

of about -0.001 under independence. For spatially cross-fitted XGBoost it is 0.59. The trees recover part of the field through country-identifying national covariates, which is exactly the leakage that makes in-sample residuals understate the spatial structure (Section 3.7). Second, the field estimated on top of the cross-fitted offset recovers the truth well (correlation between $\hat{w}$ and w of 0.81 when u is included), and its hyperparameters are close to the truth: $\hat{\rho}_w =$ 373 km against 400 and $\hat{\sigma}_w = 0.94$ against 1. Third, modelling the residual field improves forecasts for the same places. The hybrid model improves on the original XGBoost in log score in all 20 replicates (mean difference -0.0031, SE 0.0003) and in AUC-PR in 19 of 20 (mean difference $+0.011$, SE 0.001). The correctly specified parametric model is better still, in every replicate. The persistence baseline reaches an AUC-PR of 0.10, against 0.27 for the true probabilities; in these simulations persistence is a much weaker benchmark than it is likely to be in the data, where some cells record events almost every week.

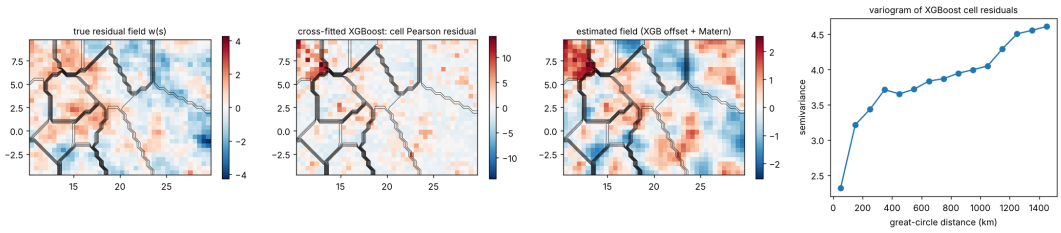


FIG 1. *S1, one replicate. From left: the true field w; standardised cell residuals of the cross-fitted XGBoost model; posterior mean of w in the hybrid model (3.5); semivariogram of the XGBoost cell residuals. Lines are country borders.*

TABLE 6

S2: estimation of the effect α of a national indicator (30 replicates per scenario). Coverage of nominal 95% intervals and the proportion of replicates in which the interval excludes zero. †Least-squares slope of SHAP values on Z_0; 95% block-bootstrap percentile interval.

Estimator	Mean	Emp. SD	Mean SE	Coverage	Reject $\alpha = 0$
No confounding, $\alpha = 0$					
Logistic regression	-0.013	0.102	0.016	0.27	0.73
+ country effects u	-0.002	0.124	0.132	0.97	0.03
+ spatial field w	0.002	0.068	0.055	0.90	0.10
+ u + w	-0.002	0.070	0.084	1.00	0.00
spatial+ with u + w	-0.002	0.077	0.085	1.00	0.00
XGBoost SHAP slope†	0.002	0.088	0.070	0.60	0.40
Confounding, $\alpha = 0$					
Logistic regression	0.469	0.095	0.017	0.00	1.00
+ country effects u	0.517	0.115	0.086	0.00	1.00
+ spatial field w	0.197	0.072	0.060	0.13	0.87
+ u + w	0.204	0.059	0.081	0.20	0.80
spatial+ with u + w	0.151	0.065	0.084	0.50	0.50
XGBoost SHAP slope†	0.394	0.093	0.086	0.00	1.00
Confounding, $\alpha = -0.3$					
Logistic regression	0.250	0.102	0.017	0.00	1.00
+ country effects u	0.269	0.117	0.085	0.00	0.77
+ spatial field w	-0.091	0.074	0.060	0.17	0.33
+ u + w	-0.086	0.059	0.082	0.17	0.20
spatial+ with u + w	-0.133	0.059	0.084	0.40	0.23
XGBoost SHAP slope†	0.178	0.093	0.075	0.00	0.70

4.4. *S2: national indicators under spatial confounding.* Three scenarios with 25 countries and 30 replicates each are considered: no confounding with $\alpha = 0$ ($\varrho = 0$); confounding with $\alpha = 0$ ($\varrho = 0.8$); and confounding with a true effect $\alpha = -0.3$. We compare the estimators of Section 3.6 with the XGBoost summary of the original analysis. For the latter, the effect of Z_0 is summarised by the least-squares slope of its SHAP values on Z_0, with intervals from 20 replicates of the 5° block bootstrap on a fixed 30% subsample of the training data, as in the original analysis.

Table 6 reports the results. Without confounding, all estimators are centred on zero, but intervals that ignore the country-level structure of the covariate are badly anticonservative. Logistic regression reports a standard error of 0.017 against an empirical SD of 0.10, so it covers in 27% of replicates and declares a non-existent effect significant in 73%. Country effects alone restore nominal coverage. The block-bootstrap summary of the original analysis

does little better than logistic regression: its 95% interval excludes zero in 40% of replicates and its interquartile range, the criterion used in the original analysis, in 70%, although Z_0 has no effect. The reason is that 5° blocks do not resample countries, the units at which Z_0 varies.

Under confounding with $\alpha = 0$, no estimator is unbiased. Logistic regression and the country-effects model report $\hat{\alpha} \approx 0.47$–$0.52$ and always reject. The spatial field removes about 60% of the bias ($\hat{\alpha} \approx 0.20$), and spatial+ removes about 70% (0.15; coverage 0.50). XGBoost reports a SHAP slope of 0.39, with an interquartile range that excludes zero in every replicate. With a true protective effect ($\alpha = -0.3$), the naive estimators and XGBoost report effects of the *wrong sign* ($+0.25$ and $+0.18$, the latter with an interquartile range excluding zero in 90% of replicates). The spatial estimators recover the sign but only a third to a half of the magnitude (-0.09 to -0.13). The lesson for the national sustainability indicators is that the data can reveal whether an association survives spatial adjustment, but not its causal size. A large SHAP contribution of a national indicator, stable across block-bootstrap replicates, is fully compatible with no effect at all.

4.5. *S3: state dependence versus persistent heterogeneity.* Two data-generating processes on a smaller 16×14 domain (224 cells, 6 countries, 104 training weeks, 20 replicates each) produce similar autocorrelation in Y_{st} for different reasons: *state dependence* with $\delta = 1$ and a weak field ($\sigma_w = 0.3$), and *heterogeneity* with no state dependence ($\delta = 0$) but a strong static field ($\sigma_w = 1.3$) and a space-time AR(1) field ($\sigma_v = 0.7$, $\phi = 0.9$ per week, $\rho_v = 300$ km). We fit model (3.1) with lags only, with lags and w, and with lags, w and v, where v is resolved at four-week steps. We also report the share of total mean absolute SHAP value that XGBoost assigns to the lag feature. Results are in Table 7.

Under state dependence, ignoring the latent field inflates $\hat{\delta}$ by about a third (1.32; coverage 0.35). Both latent specifications are approximately unbiased (1.02 and 0.98) and have nominal coverage. Under heterogeneity the contrast is much sharper. With lags only, $\hat{\delta} = 2.13$ although the true value is zero, and adding the static field halves the bias to 0.70 without restoring coverage: slowly varying episodes act as state dependence. Only the model with the space-time field brings the estimate close to zero (0.11, with an empirical SD of 0.12 and a coverage of 0.80). The remaining bias reflects the approximation of a weekly AR(1) process by a field resolved at four-week steps. In the application, the temporal resolution of v should therefore be chosen by marginal likelihood. Finally, the XGBoost importance of the lag feature is *larger* under heterogeneity (share 0.17) than under state dependence (0.11). Variable importance of recent history therefore says nothing about whether violence begets violence.

4.6. *S4: biome-specific climate effects.* The temperature effect follows (3.4), with biome means $\boldsymbol{\gamma} = (-0.3, -0.1, 0.1, 0.3)$ and a within-biome Matérn deviation η ($\sigma_\eta = 0.15$, $\rho_\eta = 600$ km); $\sigma_w = 0.8$ and 30 replicates. We compare fixed biome-specific slopes without and with $u + w$, the SVC model (3.4), the per-biome least-squares slope of the XGBoost SHAP values, and the statistic of the original analysis, the median SHAP value within each biome, with the interquartile range over 10 block-bootstrap replicates. Coverage is reported for both the biome mean γ_b and the realised biome-average effect $\bar{\beta}_b$ (Section 3.5).

The SVC model is the only estimator with close to nominal coverage for the biome means γ_b (0.90–0.97; Table 8). Its intervals are conservative for the realised biome averages $\bar{\beta}_b$, as expected, because they also account for the sampling of regions within biomes. Fixed biome slopes are nearly unbiased, but their intervals ignore heterogeneity within biomes and cover γ_b in only 37–67% of replicates, and $\bar{\beta}_b$ in 50–87%. The price of honest intervals is power: the SVC model detects the small effects ($|\gamma_b| = 0.1$) in only 20–43% of replicates. The SVC

TABLE 7

S3: estimation of the state-dependence parameter δ (20 replicates per scenario), and the share of the XGBoost mean absolute SHAP value assigned to the lag feature.

Estimator	Mean $\hat{\delta}$	Emp. SD	Mean SE	Coverage
State dependence ($\delta = 1$)				
Lags only	1.322	0.181	0.133	0.35
Lags + w	1.022	0.136	0.136	0.95
Lags + w + v	0.975	0.143	0.137	0.95
XGBoost: share of mean |SHAP| for lags	0.108	0.032	–	–
Heterogeneity ($\delta = 0$)				
Lags only	2.132	0.425	0.071	0.00
Lags + w	0.704	0.155	0.085	0.00
Lags + w + v	0.108	0.117	0.096	0.80
XGBoost: share of mean |SHAP| for lags	0.168	0.048	–	–

TABLE 8

S4: estimation of biome-specific temperature effects (30 replicates). "Detects sign": proportion of replicates in which the interval excludes zero on the correct side. ‡Median SHAP value within the biome with the interquartile range of the bootstrap replicates, as in the original analysis; it estimates neither γ_b nor $\bar{\beta}_b$.

Estimator / biome	Mean est.	Cov. γ_b	Cov. $\bar{\beta}_b$	Detects sign
Biome slopes, no latent effects				
biome 1 ($\gamma = -0.3$)	-0.263	0.47	0.60	1.00
biome 2 ($\gamma = -0.1$)	-0.089	0.67	0.77	0.57
biome 3 ($\gamma = +0.1$)	0.124	0.57	0.57	0.73
biome 4 ($\gamma = +0.3$)	0.295	0.40	0.50	0.97
Biome slopes + u + w				
biome 1 ($\gamma = -0.3$)	-0.289	0.63	0.73	1.00
biome 2 ($\gamma = -0.1$)	-0.093	0.63	0.87	0.67
biome 3 ($\gamma = +0.1$)	0.131	0.57	0.67	0.73
biome 4 ($\gamma = +0.3$)	0.319	0.37	0.80	0.97
SVC: biome means + η + u + w				
biome 1 ($\gamma = -0.3$)	-0.301	0.97	1.00	0.87
biome 2 ($\gamma = -0.1$)	-0.096	0.93	1.00	0.20
biome 3 ($\gamma = +0.1$)	0.124	0.90	1.00	0.43
biome 4 ($\gamma = +0.3$)	0.321	0.90	1.00	0.87
XGBoost SHAP slope				
biome 1 ($\gamma = -0.3$)	-0.092	0.03	0.07	0.57
biome 2 ($\gamma = -0.1$)	-0.022	0.47	0.57	0.23
biome 3 ($\gamma = +0.1$)	0.063	0.67	0.60	0.43
biome 4 ($\gamma = +0.3$)	0.113	0.07	0.07	0.53
XGBoost median SHAP‡				
biome 1 ($\gamma = -0.3$)	-0.043	0.00	0.00	0.97
biome 2 ($\gamma = -0.1$)	-0.042	0.00	0.07	1.00
biome 3 ($\gamma = +0.1$)	-0.043	0.00	0.00	0.00
biome 4 ($\gamma = +0.3$)	-0.051	0.00	0.00	0.00

model also recovers the coefficient surface (Table 9): the correlation between $\hat{\beta}(s)$ and $\beta(s)$ is 0.92, the RMSE is a third lower than with biome slopes only (0.113 against 0.172), and $\hat{\sigma}_\eta = 0.147$ against a true 0.15.

TABLE 9
S4: recovery of the spatially varying coefficient $\beta(s)$ (mean and SD over 30 replicates).

Quantity	Mean	SD
$\mathrm{cor}\{\hat{\beta}(s), \beta(s)\}$, SVC	0.917	0.043
RMSE of $\hat{\beta}(s)$, SVC	0.113	0.014
RMSE of $\hat{\beta}(s)$, biome slopes only	0.172	0.023
$\hat{\sigma}_\eta$ [truth 0.15]	0.147	0.025
$\hat{\rho}_\eta$ (km) [truth 600]	658.731	293.333

The XGBoost summaries are much less informative. The per-biome SHAP slopes have the right ordering but are attenuated to about a third of the true effects, a consequence of the shrinkage built into boosting. The median SHAP value within a biome, the statistic of the original analysis, is between -0.05 and -0.04 in *every* biome, whatever the true effect. Its bootstrap interquartile range nevertheless excludes zero in nearly every replicate, so applying the original criterion would report a "negative contribution of temperature" in all four biomes, including the two where the true effect is positive. As argued in Section 3.5, the median SHAP value describes where a biome's covariate values lie relative to the model baseline, not the strength of the effect.

5. Application to ACLED 2018–2024. All models are trained on 2018–2022 and evaluated on 2023–2024. The boosting models use the features and hyperparameters of Menicali, Marcantonio and Castruccio (2026). The computations were run on the Notre Dame Center for Research Computing.

5.1. *Reconstructing the cell-week table.* The table built by the original pipeline has 70,033,444 rows but only 34,671,180 distinct cell-weeks, so 50.5% of its rows are copies. One partition shows the pattern: 613,459 cell-weeks appear once, while others appear 2, 3, 8, 16, 32, 48 and up to 648 times. These multiplicities are products of small integers. The national indicators and IPC regions are attached by spatial intersection and then merged on cell and year, so a cell that touches several countries is multiplied at every join, and each copy combines indicators from different countries. We rebuilt the table with one row per cell-week, as described in Section 2.1. Excluding rock-and-ice cells leaves 59,344 cells and 366 weeks, or 21,719,904 cell-weeks. Of these, 4,787,224 fall in the test period, in 46,031 cells with complete climate records. The prevalence of event weeks in the test period is 1.62% for conflict and 2.73% for unrest. In the duplicated table it is 2.98% and 3.58%, because the copied border cells have more events than average.

5.2. *The boosting analysis before and after the repair.* Table 10 reproduces the XGBoost analysis on the duplicated table, then repeats it on the repaired table. In the corrected version the number of trees is chosen on 2022 rather than on the test years. On the duplicated table we recover the published performance: AUC-PR is 0.934 for conflict and 0.917 for unrest, against 0.93 and 0.91 reported. Split by cell type, however, nearly all of it comes from the duplicated border cells, with AUC-PR 0.97. On interior cells AUC-PR is 0.50–0.53.

The reason is a feature-construction artefact. The four-week event counts are computed row by row within each cell. When a cell-week has copies, the rows just before a given row are copies of the same week, so for duplicated cells the "past" counts contain the current week's events, which is the outcome. On the repaired table AUC-PR falls to 0.549 for conflict and 0.553 for unrest, with no difference between border and interior cells. A persistence baseline that ranks cell-weeks by their past four-week event count reaches 0.511 and 0.500. The covariates therefore add about 0.04–0.05 AUC-PR to recent event history. They are not the dominant source of skill implied by the published ranking.

TABLE 10

XGBoost on the 2023–2024 test period, as published (duplicated table, early stopping on the test years) and corrected (one row per cell-week, early stopping on 2022). AUC-PR is also reported separately for cells that were duplicated in the original table ("border") and for the others ("interior").

Pipeline	Outcome	Test rows	Prevalence	AUC-PR	Border cells	Interior cells	Persistence
As published	Conflict	14.43M	2.98%	0.934	0.966	0.498	0.890
As published	Unrest	14.43M	3.58%	0.917	0.968	0.528	0.852
Corrected	Conflict	4.79M	1.62%	0.549	0.588	0.528	0.511
Corrected	Unrest	4.79M	2.73%	0.553	0.542	0.555	0.500

TABLE 11

Residual diagnostics of the spatially cross-fitted XGBoost model and hyperparameters of the hybrid model, Africa and the Middle East.

Outcome	Moran's I	Neighbour residual corr.		$\hat{\rho}_w$ (km)	$\hat{\sigma}_w$	$\hat{\sigma}_u$
		within country	across border			
Conflict	0.452	0.482	0.305	277	1.24	0.35
Unrest	0.110	0.114	0.108	72	1.11	0.59

5.3. *Residual spatial structure and the hybrid model.* We fit the hybrid model (3.5) to Africa and the Middle East (longitude 20°W to 65°E, latitude 36°S to 45°N; 13,719 cells). It combines a spatially cross-fitted XGBoost offset, country effects and a Matérn field with sea cells as barriers (Remark 1). The offset's residuals have strong spatial structure for conflict (Table 11): Moran's I of the cell residuals is 0.45 ($p = 0.005$), and the semivariogram rises over the first 300–400 km before levelling off (Figure 3). Neighbouring cells in the same country have more strongly correlated residuals (0.48) than neighbours across a border (0.31), which points to country-level effects in addition to the smooth field. Unrest residuals are much less structured: Moran's I is 0.11, and the within-country and across-border correlations are equal.

The estimated fields (Figure 2) are consistent with this. For conflict, the field has a range of 277 km and a standard deviation of 1.24 on the logit scale. Its positive values form contiguous regions. Averaging $\hat{w}$ over boxes, the highest values are in eastern Democratic Republic of the Congo (1.85), northern Mozambique (1.65), Syria and Iraq (1.55), Yemen (1.36), the Lake Chad basin (1.35) and the central Sahel (0.76); the Horn of Africa is weaker (0.33). In these regions conflict is more frequent than the covariates and recent history predict. The clearest negative regions are the Arabian interior (-0.73) and the western Congo basin (-0.84). The unrest field has a range of only 72 km and consists mainly of isolated cells rather than regional patterns.

Adding the latent effects improves out-of-sample prediction for both outcomes (Table 12). For conflict, the log score falls from 0.0630 to 0.0600, AUC-PR rises from 0.508 to 0.527 and AUC-ROC from 0.942 to 0.958. For unrest, the corresponding figures are 0.0672 to 0.0633, 0.522 to 0.548 and 0.946 to 0.959. [TODO: Add the like-for-like comparison with the corrected XGBoost model on the same Africa and Middle East test rows, with the cell-clustered standard error of the log-score difference (CRC Task 6). The global fit (Task 7) follows.]

5.4. *State dependence, climate and national indicators.* [TODO: Results of the parametric models (CRC Task 8): (i) the state-dependence coefficient δ with lags only, with $u + w$, and with the space-time field v (Sahel and Horn of Africa); (ii) biome means and within-biome variation of the temperature effect from the SVC model; (iii) national indicators (EPI,

TABLE 12
Out-of-sample scores, Africa and the Middle East, 2023–2024. Lower is better for the log and Brier scores.

Outcome	Model	Log score	Brier	AUC-PR	AUC-ROC
Conflict	Cross-fitted XGBoost	0.0630	0.01608	0.508	0.942
Conflict	Cross-fitted XGBoost + $u + w$	0.0600	0.01595	0.527	0.958
Conflict	Persistence	–	–	0.228	–
Unrest	Cross-fitted XGBoost	0.0672	0.01747	0.522	0.946
Unrest	Cross-fitted XGBoost + $u + w$	0.0633	0.01698	0.548	0.959
Unrest	Persistence	–	–	0.233	–

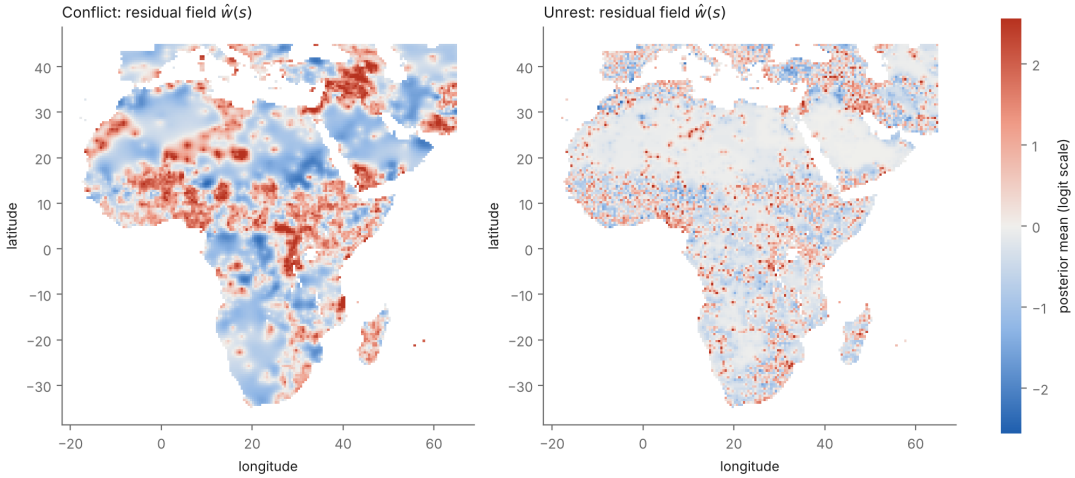


FIG 2. *Posterior mean of the residual field $w(s)$ of the hybrid model for conflict (left) and unrest (right), on the logit scale. Positive values mark persistently higher risk than predicted by the covariates, recent event history and country effects.*

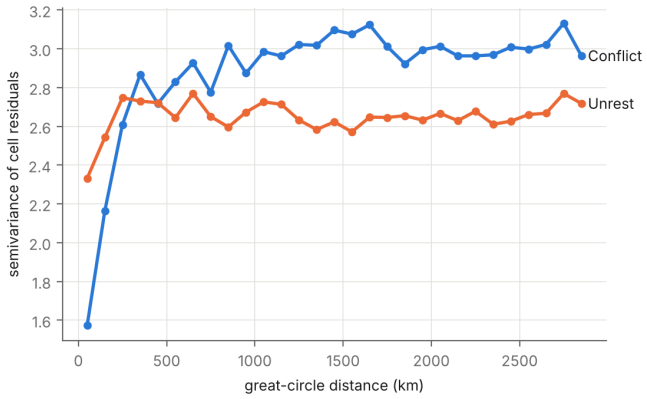


FIG 3. *Empirical semivariograms of the standardised cell residuals of the spatially cross-fitted XGBoost model, against great-circle distance.*

footprints, GHG, V-Dem) without latent effects, with $u + w$, and with spatial+, interpreted in the light of the confounding results of Section 4.4.]

6. Discussion. [TODO: Discussion (not drafted yet).]

Acknowledgments. *Use of generative AI.* Claude, a large language model developed by Anthropic, accessed through the Claude Code tool between October and November 2026, was used extensively in this work: (i) to write and test the computer code, including the latent Gaussian model and its Laplace-approximation inference engine, the simulator and the simulation studies, the reconstruction of the cell-week data set, the reproduction of the earlier boosting analysis, and the scripts that produce the tables and figures; (ii) to identify the data-processing errors in the earlier pipeline reported in Section 5; (iii) to draft and edit the text of the manuscript and to compile the bibliography; and (iv) to carry out internal adversarial reviews of the draft before submission. The authors designed and directed the analysis, checked the code, results, text and references, and take full responsibility for the content of the paper. Claude is not an author of this work. All code is available in the supplementary material.

Funding. [TODO: Funding.]

SUPPLEMENTARY MATERIAL

Code

Python implementation of the latent Gaussian model, the simulator and all simulation studies.

REFERENCES

BAKKA, H., VANHATALO, J., ILLIAN, J. B., SIMPSON, D. and RUE, H. (2019). Non-stationary Gaussian models with physical barriers. *Spatial Statistics* **29** 268–288.

CAMELETTI, M., LINDGREN, F., SIMPSON, D. and RUE, H. (2013). Spatio-temporal modeling of particulate matter concentration through the SPDE approach. *AStA Advances in Statistical Analysis* **97** 109–131.

CHEN, T. and GUESTRIN, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* 785–794.

CHERNOZHUKOV, V., CHETVERIKOV, D., DEMIRER, M., DUFLO, E., HANSEN, C., NEWEY, W. and ROBINS, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal* **21** 1–68.

DUNN, P. K. and SMYTH, G. K. (1996). Randomized quantile residuals. *Journal of Computational and Graphical Statistics* **5** 236–244.

DUPONT, E., WOOD, S. N. and AUGUSTIN, N. H. (2022). Spatial+: A novel approach to spatial confounding. *Biometrics* **78** 1279–1290.

FUGLSTAD, G. A., SIMPSON, D., LINDGREN, F. and RUE, H. (2019). Constructing priors that penalize the complexity of Gaussian random fields. *Journal of the American Statistical Association* **114** 445–452.

GELFAND, A. E., KIM, H. J., SIRMANS, C. F. and BANERJEE, S. (2003). Spatial modeling with spatially varying coefficient processes. *Journal of the American Statistical Association* **98** 387–396.

GNEITING, T. and RAFTERY, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* **102** 359–378.

HANKS, E. M., SCHLIEP, E. M., HOOTEN, M. B. and HOETING, J. A. (2015). Restricted spatial regression in practice: geostatistical models, confounding, and robustness under model misspecification. *Environmetrics* **26** 243–254.

HECKMAN, J. J. (1981). Heterogeneity and State Dependence. In *Studies in Labor Markets* (S. Rosen, ed.) 91–140. Chicago: University of Chicago Press.

HODGES, J. S. and REICH, B. J. (2010). Adding spatially-correlated errors can mess up the fixed effect you love. *The American Statistician* **64** 325–334.

KARYPIS, G. and KUMAR, V. (1998). A fast and high quality multilevel scheme for partitioning irregular graphs. *SIAM Journal on Scientific Computing* **20** 359–392.

KHAN, K. and CALDER, C. A. (2022). Restricted spatial regression methods: Implications for inference. *Journal of the American Statistical Association* **117** 482–494.

KING, G. and ZENG, L. (2001). Logistic regression in rare events data. *Political Analysis* **9** 137–163.

KRISTENSEN, K., NIELSEN, A., BERG, C. W., SKAUG, H. and BELL, B. M. (2016). TMB: Automatic differentiation and Laplace approximation. *Journal of Statistical Software* **70** 1–21.

LINDGREN, F., RUE, H. and LINDSTRÖM, J. (2011). An explicit link between Gaussian fields and Gaussian Markov random fields: the stochastic partial differential equation approach. *Journal of the Royal Statistical Society: Series B* **73** 423–498.

MENICALI, L., MARCANTONIO, R. and CASTRUCCIO, S. (2026). Detecting antecedents of violence and stanchions of peace across seasons and biomes. Unpublished manuscript.

MORAN, P. A. P. (1950). Notes on continuous stochastic phenomena. *Biometrika* **37** 17–23.

PACIOREK, C. J. (2010). The importance of scale for spatial-confounding bias and precision of spatial regression estimators. *Statistical Science* **25** 107–125.

RALEIGH, C., LINKE, A., HEGRE, H. and KARLSEN, J. (2010). Introducing ACLED: An Armed Conflict Location and Event Dataset. *Journal of Peace Research* **47** 651–660.

REICH, B. J., HODGES, J. S. and ZADNIK, V. (2006). Effects of residual smoothing on the posterior of the fixed effects in disease-mapping models. *Biometrics* **62** 1197–1206.

ROBERTS, D. R., BAHN, V., CIUTI, S., BOYCE, M. S., ELITH, J., GUILLERA-ARROITA, G., HAUENSTEIN, S., LAHOZ-MONFORT, J. J., SCHRÖDER, B., (2017). Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography* **40** 913–929.

RUE, H., MARTINO, S. and CHOPIN, N. (2009). Approximate Bayesian inference for latent Gaussian models by using integrated nested Laplace approximations. *Journal of the Royal Statistical Society: Series B* **71** 319–392.

SIGRIST, F. (2022). Gaussian process boosting. *Journal of Machine Learning Research* **23** 1–46.

SIMPSON, D., RUE, H., RIEBLER, A., MARTINS, T. G. and SØRBYE, S. H. (2017). Penalising model component complexity: A principled, practical approach to constructing priors. *Statistical Science* **32** 1–28.

WOOLDRIDGE, J. M. (2005). Simple solutions to the initial conditions problem in dynamic, nonlinear panel data models with unobserved heterogeneity. *Journal of Applied Econometrics* **20** 39–54.

ZIMMERMAN, D. L. and VER HOEF, J. M. (2022). On deconfounding spatial confounding in linear models. *The American Statistician* **76** 159–167.